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Electric circuit breaker device

Abstract

An electric circuit breaker device includes: an igniter provided to a housing; a projectile disposed in an accommodating space formed in the housing, the projectile being to be projected along the accommodating space by energy received from the igniter; a conductor piece that is provided to the housing, forms a portion of an electric circuit, and includes in a portion thereof a cutoff portion disposed crossing the accommodating space and to be cut off by the projectile; and an arc-extinguishing region that is provided in the accommodating space and in which a coolant material is disposed, the arc-extinguishing region being configured to receive the cutoff portion after being cut off. The projectile includes a first projectile configured to cut off the cutoff portion from the conductor piece by being projected by the energy received from the igniter, and a second projectile configured to press, into the arc-extinguishing region, the cutoff portion cut off by the first projectile.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to an electric circuit breaker device.

BACKGROUND ART

(2) An electric circuit may be provided with a breaker device configured to be actuated when an abnormality occurs in a device constituting the electric circuit or when an abnormality occurs in a system in which the electric circuit is mounted, thereby urgently interrupting the continuity of the electric circuit. Electric circuit breaker devices have been proposed in which, according to one aspect thereof, a projectile is moved at high speed by energy applied from an igniter or the like to forcibly and physically cut a conductor piece that forms a portion of an electric circuit (refer to Patent Documents 1 and 2 and the like, for example). Further, in recent years, electric circuit breaker devices applied to electric vehicles equipped with a high-voltage power source are becoming increasingly important.

CITATION LIST

Patent Literature

(3) Patent Document 1: JP 2014-49300 A Patent Document 2: JP 2018-6082 A

SUMMARY OF INVENTION

Technical Problem

(4) In an electric circuit breaker device, an arc is likely to occur when a conductor piece forming a portion of an electric circuit is cut. When an arc occurs, the electric circuit cannot be interrupted quickly, and thus the electric circuit breaker device must quickly extinguish the generated arc.

(5) The technique of the present disclosure has been made in view of the circumstances described above, and an object thereof is to provide an electric circuit breaker device capable of quickly extinguishing an arc during actuation.

Solution to Problem

(6) To solve the problems described above, in an electric circuit breaker device according to the present disclosure, a projectile to be projected along an accommodating space formed in a housing by energy received from an igniter includes a first projectile configured to cut off a cutoff portion from a conductor piece by being projected by the energy received from the igniter, and a second projectile configured to press, into an arc-extinguishing region of the accommodating space in which a coolant material is disposed, the cutoff portion cut off by the first projectile.

(7) More specifically, the electric circuit breaker device according to the present disclosure includes: an igniter provided to a housing; a projectile disposed in an accommodating space, the accommodating space being formed in the housing and extending in one direction, the projectile being to be projected along the accommodating space by energy received from the igniter; a conductor piece that is provided to the housing, forms a portion of an electric circuit, and includes in a portion thereof a cutoff portion disposed crossing the accommodating space and to be cut off by the projectile; and an arc-extinguishing region that is provided in the accommodating space and in which a coolant material is disposed, the arc-extinguishing region being configured to receive the cutoff portion after being cut off, in which the projectile includes a first projectile configured to cut off the cutoff portion from the conductor piece by being projected by the energy received from the igniter, and a second projectile configured to press, into the arc-extinguishing region, the cutoff portion cut off by the first projectile.

(8) Here, the second projectile may be attached to the first projectile prior to actuation of the igniter, and may be projected from the first projectile by the energy received from the igniter.

- (9) Further, the second projectile may be smaller in size than the first projectile.
- (10) Further, the second projectile may be smaller in transverse cross-sectional area than the first projectile.
- (11) Further, the second projectile may be attached to the first projectile with the second projectile positioned coaxially with the first projectile prior to actuation of the igniter.
- (12) Further, the second projectile may be attached to the first projectile with a center axis of the second projectile extending through or near a planar center portion of the cutoff portion prior to actuation of the igniter.
- (13) Further, the first projectile may include a cutoff surface disposed facing the cutoff portion prior to actuation of the igniter and configured to cut off the cutoff portion, an attachment recessed portion opening in the cutoff surface and configured to be attached with the second projectile, and a communication path through which the energy received from the igniter is guided to a pressure receiving portion of the second projectile attached to the attachment recessed portion.
- Advantageous Effects of Invention
- (14) According to the present disclosure, it is possible to provide an electric circuit breaker device capable of quickly extinguishing an arc that occurs during actuation.
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Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a view illustrating an internal structure of a breaker device.
- (2) FIG. 2 is a top view of a conductor piece.
- (3) FIG. 3 is an exploded view of a projectile 30.
- (4) FIG. 4A is a view illustrating a middle of actuation situation of the breaker device.
- (5) FIG. 4B is a view illustrating an actuation completed situation of the breaker device.
- (6) FIG. 5 is a view illustrating a modification of the breaker device.

DESCRIPTION OF EMBODIMENTS

(7) An electric circuit breaker device according to an embodiment of the present disclosure will be described below with reference to the drawings. Note that the configurations, combinations thereof, and the like in the embodiment are examples, and various additions, omissions, substitutions, and other changes may be made as appropriate without departing from the spirit of the present disclosure. The present disclosure is not limited by the embodiments and is limited only by the claims.

(8) Configuration

- (9) FIG. 1 is a view illustrating an internal structure of an electric circuit breaker device (hereinafter simply referred to as the “breaker device”) 1 according to an embodiment. The breaker device 1 is a device configured to interrupt an electric circuit included in a vehicle, an electric home appliance, or the like when an abnormality occurs in the electric circuit or in a system including a lithium battery (lithium ion battery, for example) of the electric circuit, thereby preventing great damage, for example. In the present specification, a cross section along the height direction illustrated in FIG. 1 (direction in which an accommodating space 13 described later extends) is referred to as a vertical cross section of the breaker device 1, and a cross section in a direction orthogonal to the vertical cross section is referred to as a transverse cross section of the breaker device 1. FIG. 1 illustrates a state prior to actuation of the breaker device 1.
- (10) The breaker device 1 includes a housing 10 as an outer shell member, an igniter 20, a projectile 30, a conductor piece 50, and a coolant material. The housing includes the accommodating space 13 that extends in a direction from a first end portion 11 on an upper end side to a second end portion 12 on a lower end side. This accommodating space 13 is a space formed in a straight line, making the projectile 30 movable, and extends along a vertical direction

of the breaker device **1** As illustrated in FIG. 1, the accommodating space **13** formed inside the housing **10** accommodates the projectile **30**. Note that as described in detail later, the projectile **30** includes a first projectile **40** and a second projectile **70** attached to the first projectile **40** in a pre-actuation initial state prior to actuation of the breaker device **1**. However, in the present specification, the vertical direction of the breaker device **1** merely indicates a relative positional relationship among the elements in the breaker device **1** for convenience of description of the embodiment.

(11) Housing

(12) The housing **10** includes a housing body **100**, a top holder **110**, and a bottom container **120**. The housing body **100** is bonded to the top holder **110** and the bottom container **120**, thereby forming the housing **10** that is integral.

(13) The housing body **100** has, for example, a substantially prismatic outer shape. However, the shape of the housing body **100** is not particularly limited. The housing body **100** includes a cavity portion formed therethrough along the vertical direction. This cavity portion forms a portion of the accommodating space **13**. Furthermore, the housing body **100** includes an upper surface **101** to which a flange portion **111** of the top holder **110** is fixed and a lower surface **102** to which a flange portion **121** of the bottom container **120** is fixed. In the present embodiment, an upper tubular wall **103** having a tubular shape is provided erected upward from the upper surface **101** on the outer circumferential side of the upper surface **101** in the housing body **100**. In the present embodiment, the upper tubular wall **103** has a rectangular tubular shape, for example, but may have other shapes. On the outer circumferential side of the lower surface **102** in the housing body **100**, a lower tubular wall **104** having a tubular shape is provided suspended downward from the lower surface **102**. In the present embodiment, the lower tubular wall **104** has a rectangular tubular shape, for example, but may have other shapes. The housing body **100** configured as described above can be formed from an insulating member such as a synthetic resin, for example. For example, the housing body **100** may be formed from nylon, which is a type of polyamide synthetic resin.

(14) Top Holder

(15) Next, the top holder **110** will be described. The top holder **110** is, for example, a cylindrical member having a stepped cylindrical tubular shape with a hollow inside. The top holder **110** includes a small diameter cylinder portion **112** positioned on the upper side (first end portion **11** side), a large diameter cylinder portion **113** positioned on the lower side, a connection portion **114** connecting these, and the flange portion **111** extending outward from a lower end of the large diameter cylinder portion **113**. For example, the small diameter cylinder portion **112** and the large diameter cylinder portion **113** are coaxially disposed and have cylindrical tubular shapes. The large diameter cylinder portion **113** has a diameter slightly larger than that of the small diameter cylinder portion **112**. The connection portion **114** extends in a radial direction of the small diameter cylinder portion **112** and the large diameter cylinder portion **113**, thereby connecting them to each other.

(16) The contour of the flange portion **111** in the top holder **110** has a substantially quadrangular shape that fits inside the upper tubular wall **103** in the housing body **100**. For example, the flange portion **111** may be integrally fastened to the upper surface **101** in the housing body **100** using a screw or the like, or may be fixed thereto by a rivet or the like, in a state of being disposed inside the upper tubular wall **103**. Further, the top holder **110** may be bonded to the housing body **100** in a state where a sealant is applied between the upper surface **101** of the housing body **100** and a lower surface of the flange portion **111** in the top holder **110**. This can increase airtightness of the accommodating space **13** formed in the housing **10**. Further, instead of the sealant or in combination with the sealant, an O-ring may be interposed between the upper surface **101** of the housing body **100** and the flange portion **111** of the top holder **110** to increase the airtightness of the accommodating space **13**.

(17) The cavity portion formed inside the small diameter cylinder portion **112** in the top holder **110** functions as an accommodating space for accommodating a portion of the igniter **20** as illustrated

in FIG. 1. Further, the cavity portion formed inside the large diameter cylinder portion **113** in the top holder **110** communicates with the cavity portion of the housing body **100** positioned below, and forms a portion of the accommodating space **13**. The top holder **110** configured as described above can be formed from an appropriate metal member, such as stainless steel or aluminum, having excellent strength and durability, for example. However, a material for forming the top holder **110** is not particularly limited. Also, for the shape of the top holder **110**, the above aspect is an example and other shapes may be adopted.

(18) Bottom Container

(19) Next, the bottom container **120** will be described. The bottom container **120** has a substantially tubular bottomed shape with a hollow inside, and includes a side wall portion **122**, a bottom wall portion **123** connected to a lower end of the side wall portion **122**, and a flange portion **121** connected to an upper end of the side wall portion **122**. The side wall portion **122** has, for example, a cylindrical tubular shape. The flange portion **121** extends outward from the upper end of the side wall portion **122**. The contour of the flange portion **121** in the bottom container **120** has a substantially quadrangular shape that fits inside the lower tubular wall **104** in the housing body **100**. For example, the flange portion **121** may be integrally fastened to the lower surface **102** in the housing body **100** using a screw or the like, or may be fixed thereto by a rivet or the like, in a state of being disposed inside the lower tubular wall **104**. Here, the bottom container **120** may be bonded to the housing body **100** in a state where the sealant is applied between the lower surface **102** of the housing body **100** and an upper surface of the flange portion **121** in the bottom container **120**. This can increase airtightness of the accommodating space **13** formed in the housing **10**. Further, instead of the sealant or in combination with the sealant, an O-ring may be interposed between the lower surface **102** of the housing body **100** and the flange portion **121** of the bottom container **120** to increase the airtightness of the accommodating space **13**.

(20) Note that the above aspect regarding the shape of the bottom container **120** is an example, and other shapes may be adopted. Further, the cavity portion formed inside the bottom container **120** communicates with the housing body **100** positioned above, and forms a portion of the accommodating space **13**. The bottom container **120** configured as described above can be formed from an appropriate metal member, such as stainless steel or aluminum, having excellent strength and durability, for example. However, a material for forming the bottom container **120** is not particularly limited. Further, the bottom container **120** may have a multilayer structure. For example, in the bottom container **120**, an exterior portion facing the outside may be formed from an appropriate metal member, such as stainless steel or aluminum, having excellent strength and durability, and an interior portion facing the accommodating space **13** may be formed from an insulating member such as a synthetic resin. Of course, the entire bottom container **120** may be formed from an insulating member.

(21) As described above, the housing **10** in the embodiment includes the housing body **100**, the top holder **110**, and the bottom container **120** that are integrally assembled, and the accommodating space **13** extending in the direction from the first end portion **11** to the second end portion **12** is formed inside the housing **10**. The accommodating space **13** accommodates the igniter **20**, the projectile **30**, a cutoff portion **53** in the conductor piece **50**, the first coolant material **60**, and the second coolant material **70** that are described below in detail.

(22) Igniter

(23) Next, the igniter **20** will be described. The igniter **20** is an electric igniter that includes an ignition portion **21** with an ignition charge, and an igniter body **22** including a pair of conduction pins (not illustrated) connected to the ignition portion **21**. The igniter body **22** is surrounded by an insulating resin, for example. Further, tip end sides of the pair of conduction pins in the igniter body **22** are exposed to the outside, and are connected to a power source when the breaker device **1** is used.

(24) The igniter body **22** includes a body portion **221** having a substantially cylindrical shape and

accommodated inside the small diameter cylinder portion **112** in the top holder **110**, and a connector portion **222** positioned on the body portion **221**. The igniter body **22** is fixed to the small diameter cylinder portion **112** by, for example, the body portion **221** being pressed to an inner circumferential surface of the small diameter cylinder portion **112**. Further, a constricted portion having an outer circumferential surface recessed as compared with other locations is annularly formed along a circumferential direction of the body portion **221** at an axially intermediate portion of the body portion **221**. An O-ring **223** is fitted into this constricted portion. The O-ring **223** is formed from, for example, rubber (silicone rubber, for example) or a synthetic resin, and functions to increase airtightness between the inner circumferential surface in the small diameter cylinder portion **112** and the body portion **221**.

(25) The connector portion **222** in the igniter **20** is disposed protruding to the outside through an opening **112A** formed at an upper end of the small diameter cylinder portion **112**. The connector portion **222** has, for example, a cylindrical tubular shape covering a side of the conduction pin, allowing connection with a connector of a power source.

(26) As illustrated in FIG. **1**, the ignition portion **21** of the igniter **20** is disposed facing the accommodating space **13** (more specifically, the cavity portion formed inside the large diameter cylinder portion **113**) of the housing **10**. The ignition portion **21** is configured as a form accommodating an ignition charge in an igniter cup, for example. For example, the ignition charge is accommodated in the igniter cup in the ignition portion **21** in a state of being in contact with a bridge wire (resistor) suspended coupling the base ends of the pair of conduction pins to each other. As the ignition charge, for example, zirconium-potassium perchlorate (ZPP), zirconium-tungsten-potassium perchlorate (ZWPP), titanium hydride-potassium perchlorate (THPP), lead tricinates, or the like may be adopted.

(27) In actuation of the igniter **20**, when an actuating current for igniting the ignition charge is supplied from the power source to the conduction pins, the bridge wire in the ignition portion **21** generates heat, and as a result, the ignition charge in the igniter cup is ignited and burns, generating a combustion gas. Then, the pressure in the igniter cup increases along with the combustion of the ignition charge in the igniter cup of the ignition portion **21**, a rupture surface **21A** of the igniter cup ruptures, and the combustion gas is discharged from the igniter cup into the accommodating space **13**. More specifically, the combustion gas from the igniter cup is discharged into a recess **44** in a piston portion **41** described later of the projectile **30** disposed in the accommodating space **13**.

(28) Conductor Piece

(29) Next, the conductor piece **50** will be described. FIG. **2** is a top view of the conductor piece **50** according to the embodiment. The conductor piece **50** is a metal body having conductivity that constitutes a portion of the components of the breaker device **1** and, when the breaker device **1** is attached to a predetermined electric circuit, forms a portion of the electric circuit, and may be referred to as a bus bar. The conductor piece **50** can be formed from a metal such as copper (Cu), for example. However, the conductor piece **50** may be formed from a metal other than copper, or may be formed from an alloy of copper and another metal. Note that examples of metals other than copper included in the conductor piece **50** include manganese (Mn), nickel (Ni), and platinum (Pt).

(30) In one aspect illustrated in FIG. **2**, the conductor piece **50** is formed as an elongated flat plate piece as a whole, and includes a first connecting end portion **51** and a second connecting end portion **52** on both end sides, and the cutoff portion **53** positioned in an intermediate portion therebetween. Connection holes **51A**, **52A** are provided in the first connecting end portion **51** and the second connecting end portion **52** of the conductor piece **50**, respectively. These connection holes **51A**, **52A** are used to connect with other conductors (lead wires, for example) in the electric circuit. Note that in FIG. **1**, the connection holes **51A** and **52A** in the conductor piece **50** are not illustrated. The cutoff portion **53** of the conductor piece **50** is a portion forcibly and physically cut by the projectile **30** (first projectile **40**) to be described later in detail and is cut off from the first connecting end portion **51** and the second connecting end portion **52** when an abnormality such as

excessive current occurs in the electric circuit to which the breaker device **1** is applied. Notches (slits) **54** are formed at both ends of the cutoff portion **53** of the conductor piece **50**, making it easy to cut and cut off the cutoff portion **53**.

(31) Here, various forms of the conductor piece **50** can be adopted, and a shape thereof is not particularly limited. While, in the example illustrated in FIG. 2, surfaces of the first connecting end portion **51**, the second connecting end portion **52**, and the cutoff portion **53** form the same surface, the form is not limited thereto. For example, the conductor piece **50** may be connected such that the cutoff portion **53** is orthogonal to or inclined relative to the first connecting end portion **51** and the second connecting end portion **52**. Further, the planar shape of the cutoff portion **53** of the conductor piece **50** is not particularly limited, either. Of course, the shapes of the first connecting end portion **51** and the second connecting end portion **52** of the conductor piece **50** are not particularly limited, either. Further, the notches **54** in the conductor piece **50** can be omitted as appropriate.

(32) Here, a pair of conductor piece holding holes **105A** and **105B** are formed in the housing body **100** according to the embodiment. The pair of conductor piece holding holes **105A** and **105B** extend in a transverse cross-sectional direction orthogonal to the vertical direction (axial direction) of the housing body **100**. More specifically, the pair of conductor piece holding holes **105A** and **105B** extend in a straight line with the cavity portion (accommodating space **13**) of the housing body **100** interposed therebetween. The conductor piece **50** configured as described above is held in the housing body **100** in a state of being inserted through the pair of conductor piece holding holes **105A** and **105B** formed in the housing body **100**. In the example illustrated in FIG. 1, the first connecting end portion **51** of the conductor piece **50** is held in a state of being inserted through the conductor piece holding hole **105A**, and the second connecting end portion **52** is held in a state of being inserted through the conductor piece holding hole **105B**. In this state, the cutoff portion **53** of the conductor piece **50** is positioned in the cavity portion (accommodating space **13**) of the housing body **100**. As described above, the conductor piece **50** attached to the housing body **100** is held orthogonally to the extending direction (axial direction) of the accommodating space **13** with the cutoff portion **53** crossing the accommodating space **13**. Note that reference sign **L1** illustrated in FIG. 2 denotes an outer circumferential position of the rod portion **42** positioned above the conductor piece **50** in a state of being attached to the housing body **100** of the breaker device **1**. In the present embodiment, the conductor piece **50** is installed with the outer circumferential position **L1** of the rod portion **42** substantially overlapping the positions of the notches **54** positioned at both ends of the cutoff portion **53**. In the present embodiment, for example, since a transverse cross-sectional area of the accommodating space **13** is larger than a transverse cross-sectional area of the cutoff portion **53**, a gap is formed on the side of the cutoff portion **53**.

(33) Coolant Material Next, a coolant material **60** disposed in the accommodating space **13** in the housing **10** will be described. Here, as illustrated in FIG. 1, prior to actuation of the breaker device **1** (igniter **20**), the cutoff portion **53** of the conductor piece **50** in a state of being held in the pair of conductor piece holding holes **105A** and **105B** in the housing body **100** is horizontally laid crossing the accommodating space **13** of the housing **10**. Hereinafter, within the accommodating space **13** of the housing **10** separated by the cutoff portion **53** of the conductor piece **50**, a region (space) in which the projectile **30** is disposed is referred to as a “projectile initial arrangement region **R1**”, and a region (space) positioned on the opposite side of the projectile **30** is referred to as an “arc-extinguishing region **R2**”. Note that in the present embodiment, the transverse cross-sectional area of the accommodating space **13** is larger than the transverse cross-sectional area of the cutoff portion **53**, and a gap is formed on the side of the cutoff portion **53**. Therefore, the projectile initial arrangement region **R1** and the arc-extinguishing region **R2** in the accommodating space **13** are not completely isolated from each other by the cutoff portion **53**, but communicate with each other via the gap. Of course, depending on the shape and size of the cutoff portion **53**, the projectile initial arrangement region **R1** and the arc-extinguishing region **R2** may be completely isolated from each

other by the cutoff portion **53**.

(34) The arc-extinguishing region **R2** of the accommodating space **13** is a region (space) for receiving the cutoff portion **53** cut off by the projectile **30** projected during actuation of the breaker device **1** (igniter **20**). In this arc-extinguishing region **R2**, the coolant material **60** as an arc-extinguishing material is disposed. The coolant material **60** is a coolant material for removing thermal energy of the arc generated and the cutoff portion **53** when the projectile **30** cuts off the cutoff portion **53** of the conductor piece **50**, and cools the arc and the cutoff portion **53**, thereby suppressing arc generation during cutting off of a current or thereby extinguishing (eliminating) the generated arc.

(35) The arc-extinguishing region **R2** of the breaker device **1** has significance as a space for receiving the cutoff portion **53** cut off from the first connecting end portion **51** and the second connecting end portion **52** of the conductor piece **50** and, at the same time, as a space for effectively extinguishing the arc generated when the cutoff portion **53** is cut off. Then, in order to effectively extinguish the arc generated when the cutoff portion **53** is cut off from the conductor piece **50**, the coolant material **60** is disposed as an arc-extinguishing material in the arc-extinguishing region **R2**. As one aspect of the embodiment, the coolant material **60** is solid. The coolant material **60** is formed into a substantially disk shape, for example, and is disposed at a bottom portion of the bottom container **120**. For example, the coolant material **60** may be formed by forming a braided metal fiber into a desired shape. Examples of the metal fiber forming the coolant material **60** include an aspect in which at least any one of steel wool or copper wool is included. However, the above aspects in the coolant material **60** are examples, and the coolant material **60** is not limited to the above aspects. For example, the coolant material **60** may be powdered or granular, or may be prepared by compression-forming powder or granules. The coolant material **60** may be liquid or gel-like instead of being solid.

(36) Projectile

(37) Next, the projectile **30** will be described. The projectile **30** includes the first projectile **40** and a second projectile **70**. FIG. 3 is an exploded view of the projectile **30**, illustrating the first projectile **40** and the second projectile **70** in a state of being separated from each other. The first projectile **40** and the second projectile **70** are formed from an insulating member such as synthetic resin, for example. Further, as illustrated in FIG. 3, the second projectile **70** is smaller in size than the first projectile **40**.

(38) First, the projectile **30** will be described with reference to FIGS. 1 and 3. The first projectile **40** includes the piston portion **41** and the rod portion **42** connected to the piston portion **41**. The piston portion **41** has a substantially cylindrical shape and has an outer diameter substantially corresponding to an inner diameter of the large diameter cylinder portion **113** in the top holder **110**. For example, the diameter of the piston portion **41** may be slightly smaller than the inner diameter of the large diameter cylinder portion **113**. The shape of the piston portion **41** can be changed as appropriate according to the shape of the large diameter cylinder portion **113** and the like.

(39) The rod portion **42** of the first projectile **40** is a rod-shaped member having an outer circumferential surface smaller in diameter than the piston portion **41**, for example, and is integrally connected to a lower end side of the piston portion **41**. A lower end surface of the rod portion **42** is formed as a cutoff surface **421** for cutting off the cutoff portion **53** from the conductor piece **50** during actuation of the breaker device **1**. The cutoff surface **421** of the first projectile **40** is disposed facing the cutoff portion **53** in a state where the first projectile **40** is disposed at an initial position illustrated in FIG. 1. Here, the rod portion **42** in the present embodiment has a substantially cylindrical shape, but the shape thereof is not particularly limited. Note that, in the initial position illustrated in FIG. 1, a region on a tip end side including the cutoff surface **421** in the rod portion **42** of the first projectile **40** is positioned in the cavity portion (forming a portion of the accommodating space **13**) of the housing body **100**. The diameter of the rod portion **42** is slightly smaller than the inner diameter of an inner circumferential surface of the housing body

100, for example. The outer circumferential surface of the rod portion **42** is guided along the inner circumferential surface of the housing body **100** during actuation of the breaker device **1**.
(40) Further, a recess **44**, which is a recessed portion having a cylindrical shape, for example, is formed on an upper surface of the piston portion **41** in the first projectile **40**. The recess **44** is configured to receive the ignition portion **21**. A bottom surface of the recess **44** is formed as a first pressure receiving portion **44A** that receives energy received from the igniter **20** during actuation of the igniter **20**. Further, a constricted portion having an outer circumferential surface recessed as compared with other locations is annularly formed along a circumferential direction of the piston portion **41** at an axially intermediate portion of the piston portion **41**. An O-ring **43** is fitted into this constricted portion. The O-ring **43** is formed from, for example, rubber (silicone rubber, for example) or a synthetic resin, and functions to increase airtightness between an inner circumferential surface in the large diameter cylinder portion **113** and the piston portion **41**.
(41) An attachment recessed portion **45** for accommodating and attaching the second projectile **70** is provided on a lower end side of the first projectile **40**. This attachment recessed portion **45** is formed in an aspect in which the attachment recessed portion **45** opens in the cutoff surface **421** of the rod portion **42** in the first projectile **40**. In the example illustrated in FIGS. **1** and **3**, the attachment recessed portion **45** has a cylindrical shape. The recess **44** and the attachment recessed portion **45** of the first projectile **40** are coaxially disposed and extend through a center axis of the first projectile **40**. Furthermore, as illustrated in FIGS. **1** and **3**, a communication path **46** that connects the recess **44** and the attachment recessed portion **45** to each other (allows communication between the recess **44** and the attachment recessed portion **45**) is provided in the first projectile **40**. The communication path **46** of the first projectile **40** is formed extending through the center axis of the first projectile **40**. The communication path **46** is also disposed coaxially with both the recess **44** and the attachment recessed portion **45**.

(42) Next, the second projectile **70** will be described. The second projectile **70** is so shaped and sized as to be accommodatable in the attachment recessed portion **45** of the first projectile **40**, and is configured as a cylindrical piston form in the present embodiment. Further, as illustrated in FIG. **3**, the second projectile **70** according to the present embodiment is smaller in size than the first projectile **40** and is smaller in transverse cross-sectional area than the first projectile **40**. For example, the diameter of the second projectile **70** may be slightly smaller than the diameter of the attachment recessed portion **45** in the first projectile **40**. Further, a constricted portion having an outer circumferential surface recessed as compared with other locations is annularly formed along a circumferential direction of the second projectile **70** at an axially intermediate portion of the second projectile **70**. An O-ring **73** is fitted into this constricted portion. The O-ring **73** is formed from, for example, rubber (silicone rubber, for example) or a synthetic resin. In the example illustrated in FIGS. **1** and **3**, the O-ring **73** is disposed at two steps on the outer circumferential surface of the second projectile **70**, but the number of steps of the O-ring **73** is not particularly limited.

(43) An upper surface of the second projectile **70** is formed as a second pressure receiving portion **71** that receives energy received from the igniter **20** during actuation of the breaker device **1** (igniter **20**). Further, a lower surface of the second projectile **70** is formed as a pressing portion **72** for pressing the cutoff portion **53** cut off by the first projectile **40** into the arc-extinguishing region **R2** during actuation of the breaker device **1** (igniter **20**). Here, when the second projectile **70** is attached to the first projectile **40**, the second projectile **70** is inserted into the attachment recessed portion of the first projectile **40** from the second pressure receiving portion **71** (upper surface) side. As a result, as illustrated in FIG. **1**, the second projectile **70** is attached to the first projectile **40** with the second pressure receiving portion **71** in the second projectile **70** facing the communication path **46** and the recess **44** in the first projectile and the pressing portion **72** disposed on an open end **45A** side of the attachment recessed portion **45**. In the present embodiment, the second projectile **70** is disposed coaxially with the first projectile **40** in a state of being attached to the attachment recessed portion **45**. However, the second projectile **70** may be decentered with respect to the first

projectile **40** in a state where the second projectile **70** is attached to the first projectile **40**.

(44) Further, for example, when the second projectile **70** is attached to the attachment recessed portion **45** of the first projectile **40**, an O-ring **73** may be compressed and deformed by being sandwiched between an inner circumferential surface of the attachment recessed portion **45** and the outer circumferential surface of the second projectile **70**. Then, a repulsive force of the O-ring **73** in the compressed and deformed state may exert a holding force for suppressing falling off of the second projectile **70** from the attachment recessed portion **45** due to its own weight. Further, in the present embodiment, an axial depth of the attachment recessed portion **45** in the first projectile **40** is slightly larger than an axial length of the second projectile **70**. Further, the axial depth of the attachment recessed portion **45** in the first projectile **40** may be equal in dimension to the axial length of the second projectile **70**. This enables the second projectile **70** to be accommodated in the attachment recessed portion **45** without a lower end portion including the pressing portion **72** of the second projectile **70** protruding from the open end **45A** of the attachment recessed portion **45** in the first projectile **40**.

(45) In the pre-actuation initial state illustrated in FIG. **1**, the projectile **30** configured as described above is disposed in the projectile initial arrangement region **R1** of the accommodating space **13** in a state where the second projectile **70** is attached to the attachment recessed portion **45** of the first projectile **40**. In the example illustrated in FIG. **1**, the piston portion **41** of the first projectile **40** is positioned on the first end portion **11** side (upper end side) in the accommodating space **13**. Further, the rod portion **42** of the first projectile **40** is disposed in a state where the cutoff surface **421** is placed on the conductor piece **50**. Here, reference sign **L1** illustrated in FIG. **2** indicates an outer circumferential position of the rod portion **42** in the first projectile positioned on the conductor piece **50** in a state of being attached to the housing body **100** of the breaker device **1**. In the pre-actuation initial state of the breaker device **1**, the outer circumferential position **L1** of the rod portion **42** in the first projectile **40** substantially overlaps the positions of the notches **54** positioned at both ends of the cutoff portion **53**.

(46) Further, reference sign **L2** illustrated in FIG. **2** indicates an outer circumferential position of the second projectile **70** attached to the first projectile **40**. As illustrated in FIG. **2**, in the pre-actuation initial state of the breaker device **1**, the second projectile **70** in the state of being attached to the first projectile **40** is provided with at least a portion of a planar region surrounded by the outer circumferential position **L2** overlapping at least a portion of a planar region of the cutoff portion **53**. More specifically, prior to actuation of the igniter **20**, the second projectile **70** is attached to the first projectile **40** with its center axis **C1** extending through or near a center position of the cutoff portion **53**. Further, in the present embodiment, the axial depth of the attachment recessed portion **45** in the first projectile **40** is slightly larger than the axial length of the second projectile **70**. Therefore, in the pre-actuation initial state of the breaker device **1**, the pressing portion **72** of the second projectile **70** is disposed slightly retracted from the cutoff surface **421** of the first projectile **40** with the cutoff portion **53** as a reference. As a result, a gap is formed between the pressing portion **72** and the cutoff portion **53**.

(47) Operation

(48) Next, operation content when the breaker device **1** is actuated to interrupt the electric circuit will be described. FIGS. **4A** and **4B** are views illustrating an actuation situation of the breaker device **1** according to the embodiment. FIG. **4A** illustrates a situation in the middle of actuation of the breaker device **1**, and FIG. **4B** illustrates a situation in which the actuation of the breaker device **1** is completed. Hereinafter, the operation content of the breaker device **1** during actuation will be described with reference to FIGS. **3**, **4A**, and **4B**.

(49) The breaker device **1** according to the present embodiment further includes an abnormality detection sensor (not illustrated) configured to detect an abnormal current of the electric circuit, and a control unit (not illustrated) configured to control the actuation of the igniter **20**. In addition to the current flowing through the conductor piece **50**, the abnormality detection sensor may be

capable of detecting a voltage and a temperature of the conductor piece **50**. Further, the control unit of the breaker device **1** is a computer capable of performing a predetermined function by executing a predetermined control program, for example. The predetermined function of the control unit may be realized by corresponding hardware. Then, when excessive current flows through the conductor piece **50** forming a portion of the electric circuit to which the breaker device **1** is applied, the abnormal current is detected by the abnormality detection sensor. Abnormality information regarding the detected abnormal current is passed from the abnormality detection sensor to the control unit. For example, the control unit is energized from an external power source (not illustrated) connected to the conduction pin of the igniter **20** and actuates the igniter based on the current value detected by the abnormality detection sensor. Here, the abnormal current may be a current value that exceeds a predetermined threshold value set for protection of a predetermined electric circuit. Note that the abnormality detection sensor and the control unit described above need not be included in the components of the breaker device **1**, and may be included in a device separate from the breaker device **1**, for example. Further, the abnormality detection sensor and the control unit are not essential components of the breaker device **1**.

(50) For example, when an abnormal current of the electric circuit is detected by an abnormality detection sensor that detects an abnormal current of the electric circuit, the control unit of the breaker device **1** actuates the igniter **20**. That is, an actuating current is supplied from the external power source (not illustrated) to the conduction pin of the igniter **20**, and as a result, the ignition charge in the ignition portion **21** is ignited and burns, generating a combustion gas. Then, the rupture surface **21A** ruptures due to rise in pressure in the ignition portion **21**, and the combustion gas of the ignition charge is discharged from the inside of the ignition portion **21** into the accommodating space **13**.

(51) As described above, the projectile **30** in the breaker device **1** includes the first projectile **40** and the second projectile **70**. The projectile **30** is configured to be projected from the initial position by receiving energy received from the igniter **20** during actuation, more specifically, energy of the combustion gas generated by combustion of the ignition charge in the ignition portion **21**, and is movable along the accommodating space **13**. The first projectile **40** and the second projectile **70** in the projectile **30** have functions (roles) different from each other. Specifically, during actuation of the breaker device **1** (igniter **20**), the first projectile **40** is projected toward the second end portion **12** side in the accommodating space **13** by the energy received from the combustion gas of the ignition charge in the igniter **20**, thereby functioning to cut off the cutoff portion **53** from the conductor piece **50**. On the other hand, during actuation of the breaker device **1** (igniter **20**), the second projectile **70** is projected toward the second end portion **12** side in the accommodating space **13** by the energy received from the combustion gas of the ignition charge in the igniter **20**, thereby functioning to press, into the arc-extinguishing region **R2**, the cutoff portion **53** cut off by the first projectile **40**. Hereinafter, the operation content of the first projectile **40** and the second projectile **70** during actuation of the breaker device **1** (igniter **20**) will be described in detail.

(52) As illustrated in FIG. **1**, the ignition portion **21** of the igniter **20** is received in the recess **44** of the piston portion **41** of the first projectile **40**, and the rupture surface **21A** of the ignition portion **21** is disposed facing the first pressure receiving portion **44A** of the recess **44** in the first projectile **40**. Therefore, the combustion gas from the ignition portion **21** is discharged toward the recess **44** of the first projectile **40**, and the pressure (combustion energy) of the combustion gas is transmitted to the upper surface of the piston portion **41** including a first pressure receiving surface **44A**. Due to this, the upper surface of the piston portion **41** including the first pressure receiving surface **44A** in the first projectile **40** is pressed, and the first projectile **40** is vigorously biased downward (toward the second end portion **12** side). As a result, the cutoff surface **421** formed on a lower end side of the rod portion **42** in the first projectile **40** is strongly pressed against the boundary portions (the portions where the notches **54** are formed) between the first connecting end portion **51** and the cutoff portion **53** and between the second connecting end portion **52** and the cutoff portion **53** of

the conductor piece **50**. In this manner, for example, the cutoff portion **53** of the conductor piece **50** is pressingly cut by shearing, whereby the cutoff portion **53** can be cut off from the conductor piece **50**.

(53) As illustrated in FIG. **4A**, the first projectile **40** moves downward (toward the second end portion **12** side) in the extending direction (axial direction) of the accommodating space **13** by a predetermined stroke until a lower end surface **411** of the piston portion **41** abuts (collides with) the upper surface **101** of the housing body **100**. A state where the lower end surface **411** of the piston portion **41** abuts (collides with) a stopper portion **101A** on the upper surface **101** of the housing body **100** in this manner and thereby restricts the first projectile **40** from moving further downward (toward the second end portion **12** side) is referred to as the “movement restriction state”. As illustrated in FIG. **4A**, in the breaker device **1** according to the present embodiment, the length of the rod portion **42** or the dimension in the vertical direction of the arc-extinguishing region **R2** is set so that the cutoff surface **421** of the rod portion **42** is positioned in a relatively upper region of the arc-extinguishing region **R2** when the first projectile **40** is projected from the initial position during actuation and is brought into the movement restriction state.

(54) Note that a holding portion for holding the lower end surface **411** of the piston portion **41** in a state of abutting the stopper portion **101A** when the lower end surface **411** of the piston portion **41** in the first projectile **40** collides with the stopper portion **101A** during actuation of the breaker device **1** may be provided on at least any one of the lower end surface **411** of the piston portion **41** or the stopper portion **101A**. Such a holding portion is not particularly limited. For example, the holding portion may be formed by a protrusion provided on the lower end surface **411** of the piston portion **41** or the stopper portion **101A**. For example, when the lower end surface **411** of the piston portion **41** collides with the stopper portion **101A**, a protrusion provided on the lower end surface **411** of the piston portion **41** pierces the stopper portion **101A**, or a protrusion provided on the stopper portion **101A** pierces the lower end surface **411** of the piston portion **41**, whereby the lower end surface **411** of the piston portion **41** can be held in a state of abutting the stopper portion **101A**. Alternatively, the holding portion may be formed not by actively providing the protrusion as described above, but by engagement between a round internal corner portion **47** formed at a boundary portion between the lower end surface **411** of the piston portion **41** and the outer circumferential surface of the rod portion **42** as illustrated in FIG. **1**, and a right angle external corner portion **106** formed by the stopper portion **101A** (upper surface **101**) and the inner circumferential surface of the housing body **100** connected at a right angle. In this case, when the lower end surface **411** of the piston portion **41** collides with the stopper portion **101A** during actuation of the breaker device **1**, the lower end surface **411** of the piston portion **41** may be held in a state of abutting the stopper portion **101A**, with the right angle external corner portion **106** biting (piercing) the round internal corner portion **47** to be engaged therewith.

(55) Next, the operation of the second projectile **70** during actuation of the breaker device **1** (igniter **20**) will be described. As described above, in the pre-actuation initial state of the breaker device **1**, the second projectile **70** is attached to the attachment recessed portion **45** of the first projectile **40**. As described above, the recess **44** and the attachment recessed portion **45** in the first projectile **40** communicate with each other via the communication path **46**, and the second pressure receiving portion **71** of the second projectile **70** in the state of being attached to the first projectile **40** is disposed facing the lower end of the communication path **46**. Therefore, a portion of the combustion gas from the ignition portion **21** discharged toward the recess **44** of the first projectile **40** during actuation of the breaker device **1** (igniter **20**) is guided to the second pressure receiving portion **71** of the second projectile **70** through the communication path **46**, and as a result, the pressure (combustion energy) of the combustion gas is transmitted to the second pressure receiving portion **71** of the second projectile **70**. Due to this, the second pressure receiving portion **71** of the second projectile **70** attached (accommodated) in the attachment recessed portion **45** of the first projectile **40** is pressed, and the second projectile **70** is vigorously biased downward (toward the

second end portion 12 side). As a result, the second projectile 70 stored in the attachment recessed portion 45 of the first projectile 40 protrudes downward from the open end 45A of the attachment recessed portion 45 and is projected. Due to this, the cutoff portion 53 cut off from the conductor piece 50 by the rod portion 42 of the first projectile 40 as illustrated in FIG. 4A is pressed downward by the pressing portion 72 of the second projectile 70 projected from the first projectile 40, whereby the cutoff portion 53 can be pressed into a bottom portion side (that is, the second end portion 12 side) of the arc-extinguishing region R2 as illustrated in FIG. 4B.

(56) As described above, the projectile 30 of the breaker device 1 according to the present embodiment includes the first projectile 40 and the second projectile 70, which are projected in two steps by receiving energy of the combustion gas generated by the burning of the ignition charge of the ignition portion 21 during actuation of the igniter 20. That is, when the first projectile 40, which is projected by the energy received from the combustion gas of the ignition charge during actuation of the igniter 20, is pressed down toward the second end portion 12 side of the accommodating space 13, the cutoff portion 53 is pressingly cut by the cutoff surface 421, whereby the cutoff portion 53 can be cut off from the conductor piece 50. As a result, the first connecting end portion 51 and the second connecting end portion 52 positioned at both ends of the conductor piece 50 are electrically disconnected, and the predetermined electric circuit to which the breaker device 1 is applied can be forcibly interrupted.

(57) Then, similarly to the first projectile 40, the second projectile 70 is projected from the first projectile 40 toward the second end portion 12 side by the energy received from the combustion gas of the ignition charge generated during actuation of the igniter 20. Due to this, the cutoff portion 53 can be separated from the cutoff surface 421 of the first projectile 40, for example, by the pressing portion 72 of the second projectile 70, and the cutoff portion 53 can be swiftly pressed into the bottom portion side (second end portion 12 side) of the arc-extinguishing region R2. As a result, the cutoff portion 53 pressed into the bottom portion side of the arc-extinguishing region R2 by the second projectile 70 is rapidly cooled by the coolant material 60 disposed in the arc-extinguishing region R2, whereby the arc generated when the cutoff portion 53 is cut off from the first connecting end portion 51 and the second connecting end portion 52 can be quickly extinguished. As a result, it is possible to quickly interrupt the electric circuit to which the breaker device 1 is applied in a case where an abnormality is detected in the electric circuit, or the like. That is, by effectively suppressing a prolonged extinguishing of the arc generated when the electric circuit is interrupted, it is possible to suppress a prolonged interruption of the electric circuit. Further, according to the breaker device 1, it is possible to suitably suppress the generation of a large spark or flame or the generation of a loud impact sound when the electric circuit is interrupted. Further, damage to the housing 10 and the like of the breaker device 1 caused by these can also be suppressed.

(58) As described above, according to the breaker device 1, separately from the first projectile 40 for cutting off the cutoff portion 53 from the conductor piece 50 during actuation of the igniter 20, there is included the second projectile 70, which is projected from the first projectile 40 to press the cutoff portion 53 cut off by the first projectile 40 into the bottom portion side (second end portion 12 side) of the arc-extinguishing region R2. By adopting such a two-step mechanism in the projectile 30, even if an axial length of the rod portion 42 in the first projectile 40 is designed to be short, the second projectile 70 can separate the cutoff portion 53 from the cutoff surface 421 of the first projectile 40 and press the cutoff portion 53 into the bottom portion side (second end portion 12 side) of the arc-extinguishing region R2. Due to this, the cutoff portion 53 after being cut off can be swiftly moved away from the first connecting end portion 51 and the second connecting end portion 52 in the conductor piece 50, the arc when the electric circuit is interrupted can be reduced, and the insulation performance thereof can be improved.

(59) On the other hand, in the known breaker device without the two-step projection mechanism of the projectile, in order to increase the distance between the conductor piece and the cut portion, a

movement stroke of the projectile corresponding to the distance by which the cut portion should be separated from the conductor piece is normally required, and therefore, the axial length of the projectile has to be increased in accordance with the movement stroke. On the other hand, it is sufficient for the axial length of the rod portion **42** in the first projectile **40** according to the present embodiment to have a length sufficient for cutting off the cutoff portion **53** by the rod portion **42** during actuation of the igniter **20**, and it is not necessary to press the cutoff portion **53** by the rod portion **42** into the bottom portion side of the arc-extinguishing region **R2**. For example, the axial length of the rod portion **42** in the first projectile **40** is only required to be set to such a length that when the piston portion **41** is brought into the movement restriction state during actuation of the igniter **20**, the position of the cutoff surface **421** is positioned lower than the position of the lower surface (the surface facing the arc-extinguishing region **R2**) of the cutoff portion **53** in the pre-actuation initial state. Due to this, while the axial length of the rod portion **42** in the first projectile **40** is shortened, the cutoff portion **53** can be cut off at the time of projection, and the cutoff portion **53** after being cut off can be swiftly separated away from the first connecting end portion **51** and the second connecting end portion **52**. Thus, being able to shorten the axial length of the rod portion **42** and, by extension, the axial length of the first projectile **40** has the following advantages.

(60) That is, in the pre-actuation initial state of the breaker device **1**, as illustrated in FIG. **1**, the projectile **1** is disposed in the projectile initial arrangement region **R1**, that is, above the cutoff portion **53** of the conductor piece **50** in the accommodating space **13**. Therefore, as the axial length of the first projectile **40** increases, it is necessary to increase the axial length of the projectile initial arrangement region **R1**, and it is necessary to increase the height dimension of the housing **10**. On the other hand, according to the breaker device **1** of the present embodiment, since the axial length of the first projectile **40** (rod portion **42**) can be shortened, the height dimension of the housing **10** can be reduced. As described above, according to the breaker device **1** of the present embodiment, it is possible to obtain an effect of improving the insulation performance (an effect of reducing the arc) when the electric circuit is interrupted while achieving downsizing of the entire housing **10**.

(61) Note that in the breaker device **1**, the timing at which the second projectile **70** is projected from the first projectile **40** during actuation of the igniter **20** is not particularly limited. For example, the second projectile **70** may be projected from the first projectile **40** at the moment when the cutoff portion **53** is removed by the cutoff surface **421** of the first projectile **40**, or, as illustrated in FIG. **4A**, the second projectile **70** may be projected from the first projectile **40** at a timing after the movement restriction state is reached where the lower end surface **411** of the piston portion **41** abuts (collides with) the stopper portion **101A** of the housing body **100**. Alternatively, the second projectile **70** may be projected from the first projectile **40**, after the first projectile **40** removes the cutoff portion **53** during actuation of the igniter **20**, at a timing in the process (middle) of the movement restriction state being reached.

(62) Furthermore, according to the breaker device **1**, as described above, the second projectile **70** is configured to be attached to the first projectile **40** prior to actuation of the igniter **20** (pre-actuation initial state) and projected from the first projectile **40** by the energy received from the igniter **20**. Due to this, it is possible to adopt, for the second projectile **70**, a reasonable arrangement aspect suitable for separating, from the cutoff surface **421** of the first projectile **40**, the cutoff portion **53** after being cut off and pressing the cutoff portion **53** into the bottom portion side (second end portion **12** side) of the arc-extinguishing region **R2**.

(63) Furthermore, in the present embodiment, since the second projectile **70** is configured as a projectile having a smaller transverse cross-sectional area than that of the first projectile **40**, it is possible to adopt an aspect suitable for attaching the second projectile **70** to the first projectile **40** in the pre-actuation initial state of the breaker device **1**. Further, since the second projectile **70** is smaller in size than the first projectile **40**, it is possible to reduce an impact when the cutoff portion **53** cut off during actuation of the breaker device **1** collides with the bottom wall portion **123** of the bottom container **120**. Therefore, even if the thickness of the bottom wall portion **123** in the bottom

container **120** is reduced, deformation, damage, and the like of the bottom wall portion **123** can be suppressed. However, the aspect of the second projectile **70** is not particularly limited as long as it is possible to press, into the arc-extinguishing region **R2**, the cutoff portion **53** cut off by the first projectile **40** during actuation of the igniter **20**. For example, the second projectile **70** may be disposed in a state of being spaced apart from the first projectile **40** without being attached to the first projectile **40** in the pre-actuation initial state. Further, it is not necessary for the second projectile **70** to be smaller in size than the first projectile **40**. The second projectile **70** may have a size equal to that of the first projectile **40**, or the second projectile **70** may be larger in size than the first projectile **40**.

(64) Further, according to the breaker device **1**, the second projectile **70** is attached to the first projectile **40** with the second projectile **70** positioned coaxially with the first projectile **40**. Due to this, when the second projectile **70** is projected from the first projectile **40**, the second projectile **70** can press, in a well-balanced manner, the cutoff portion **53** cut off by the first projectile **40** into the bottom portion side (second end portion **12** side) of the arc-extinguishing region **R2**. In particular, in the present embodiment, prior to actuation of the igniter **20**, the second projectile **70** is attached to the first projectile **40** with the center axis **C1** of the second projectile **70** extending through or near the planar center portion of the cutoff portion **53**. Due to this, when the second projectile **70** is projected from the first projectile **40**, the second projectile **70** can press the cutoff portion **53** cut off by the first projectile **40** at or near the planar center portion of the cutoff portion **53**, whereby the cutoff portion **53** can be smoothly pressed into the bottom portion side (second end portion **12** side) of the arc-extinguishing region **R2**. Note that as the second projectile **70** is attached to the first projectile **40** with the center axis **C1** of the second projectile **70** extending through the planar center portion of the cutoff portion **53**, the cutoff portion **53** cut off by the first projectile **40** during actuation of the igniter **20** can be further smoothly pressed into the bottom portion side (second end portion **12** side) of the arc-extinguishing region **R2** by the second projectile **70**.

(65) Further, the first projectile **40** in the present embodiment includes the cutoff surface **421** disposed facing the cutoff portion **53** prior to actuation of the igniter **20** and configured to cut off the cutoff portion **53**, the attachment recessed portion **45** opening in the cutoff surface **421** and configured to be attached with the second projectile **70**, and the communication path **46** through which the energy received from the igniter **20** is guided to the second pressure receiving portion **71** of the second projectile **70** attached to the attachment recessed portion **45**. Due to this, the combustion gas generated during actuation of the igniter **20** can be suitably introduced via the communication path **46** into the second pressure receiving portion **71** of the second projectile **70** attached to the attachment recessed portion **45** of the first projectile **40**. Then, the second projectile **70** can be smoothly projected from the first projectile **40** by the pressure (combustion energy) of the combustion gas introduced into the second pressure receiving portion **71**.

(66) Note that in the above embodiment, in the example illustrated in FIG. **4**, an aspect in which the second projectile **70** is projected to be completely separated from the first projectile **40** during actuation of the breaker device **1** has been described as an example. However, the present disclosure is not limited to this aspect. For example, as in the modification illustrated in FIG. **5**, it is possible to adopt specifications in which a portion of the second projectile **70** projected from the first projectile **40** during actuation of the breaker device **1** remains in the attachment recessed portion **45** in the first projectile **40**. FIG. **5** illustrates a situation in which the actuation of the breaker device **1** according to the modification is completed.

(67) While the embodiment of the electric circuit breaker device according to the present disclosure has been described above, each of the aspects disclosed in the present specification can be combined with any other feature disclosed in the present specification.

REFERENCE SIGNS LIST

(68) **1** Breaker device **10** Housing **13** Accommodating space **20** Igniter **30** Projectile **40** First projectile **50** Conductor piece **53** Cutoff portion **60** Coolant material **70** Second projectile

Claims

1. An electric circuit breaker device comprising: an igniter provided on a housing; a projectile disposed in an accommodating space, the accommodating space being formed in the housing and extending in one direction, the projectile being configured to be projected along the accommodating space by energy received from the igniter; a conductor piece provided on housing and forming portion of an electric circuit, the conductor piece including a cutoff portion disposed crossing the accommodating space and configured to be cut off by the projectile; and an arc-extinguishing region that is provided in the accommodating space and in which a coolant material is disposed, the arc-extinguishing region being configured to receive the cutoff portion after being cut off, wherein the projectile includes a first projectile configured to cut off the cutoff portion from the conductor piece by being projected by the energy received from the igniter, and a second projectile configured to press, into the arc-extinguishing region, the cutoff portion cut off by the first projectile.
 2. The electric circuit breaker device according to claim 1, wherein the second projectile is attached to the first projectile prior to actuation of the igniter, and is projected from the first projectile by the energy received from the igniter.
 3. The electric circuit breaker device according to claim 2, wherein the second projectile is smaller in size than the first projectile.
 4. The electric circuit breaker device according to claim 2, wherein the second projectile is smaller in a transverse cross-sectional area than the first projectile.
 5. The electric circuit breaker device according to claim 2, wherein the second projectile is attached to the first projectile with the second projectile positioned coaxially with the first projectile prior to actuation of the igniter.
 6. The electric circuit breaker device according to claim 2, wherein the second projectile is attached to the first projectile with a center axis of the second projectile extending through or near a planar center portion of the cutoff portion prior to actuation of the igniter.
 7. The electric circuit breaker device according to claim 2, wherein the first projectile includes a cutoff surface disposed facing the cutoff portion prior to actuation of the igniter and configured to cut off the cutoff portion, an attachment recessed portion opening in the cutoff surface and configured to be attached with the second projectile, and a communication path through which the energy received from the igniter is guided to a pressure receiving portion of the second projectile attached to the attachment recessed portion.
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